

RABEYA AKTER

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EDUCATION

Bachelor of Science in Robotics and Mechatronics Engineering

Jan 2019 - Mar 2024

University of Dhaka

CGPA: 3.75/4.00 (Final Year: 3.88/4.00)

Relevant Coursework:

Introduction to Machine Learning, Digital Image Processing and Robot Vision, Digital Signal Processing, Object-Oriented Programming, Fundamentals of Programming, Linear Algebra, Multivariate and Vector Calculus, Statistics for Engineers

RESEARCH INTERESTS

Reinforcement Learning, Multimodal Learning, Vision-Language Models, Embodied AI, Language Model Post-Training, Computer Vision

PUBLICATIONS

PUBLISHED

- **LAGEA: Language Guided Embodied Agents for Robotic Manipulation** [\[Paper\]](#)
Authors : Abdul Monaf Chowdhury, Akm Moshir Rahman Mozumder, **Rabeya Akter**, Safaeid Hossain Arib
Conference : [ICML 2026](#)
- **T3Time: Tri-Modal Time Series Forecasting via Adaptive Multi-Head Alignment and Residual Fusion** [\[Paper\]](#)
Authors: Abdul Monaf Chowdhury, **Rabeya Akter**, Safaeid Hossain Arib
Conference: [AAAI 2026](#)
- **SignFormer-GCN : Continuous Sign Language Translation Using Spatio-Temporal Graph Convolutional Networks** [\[Paper\]](#) [\[Extended Abstract\]](#)
Authors : Safaeid Hossain Arib[✉], **Rabeya Akter**[✉], Sejuti Rahman, Shafin Rahman (✉Equal contribution)
Journal: [PLOS One](#)
Workshop: [WiML Workshop, NeurIPS 2025](#)
- **Classical machine learning approach for human activity recognition using location data** [\[Paper\]](#) [\[Poster\]](#)
Authors: Safaeid Hossain Arib, **Rabeya Akter**, Omar Shahid, and Md Atiqur Rahman Ahad.
Conference: *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers*

PREPRINT

- **Teach Yourself Where to Look: On-Policy Attention Self-Distillation for Reasoning** [\[Paper\]](#)
Authors: Safaeid Hossain Arib, **Rabeya Akter**, Ishmam Nur Swapnil, Md Faiyaz Abdullah Sayeedi, Tasnim Mohiuddin, Md Mofijul Islam
Under review: [ICLR 2027](#)
- **EK-BANGLA: When Hand-Object Interaction Meets Low-Resource Generation**
Authors: **Rabeya Akter**, Md Salman Shamil, A. K. M. Fazley Rabbi, Arpa Deb, Ridwain Islam, Md Adnan Arefeen
Under review: [WACV 2027](#)
- **Counting Through Occlusion: Framework for Open World Amodal Counting** [\[Paper\]](#)
Authors: Safaeid Hossain Arib, **Rabeya Akter**, Abdul Monaf Chowdhury, Md Jubair Ahmed Surov, Md Mehedi Hasan
Under review: [WACV 2027](#)
- **Bornil : An open-source sign language data crowdsourcing platform for AI enabled dialect-agnostic communication** [\[Paper\]](#)
Authors: Shahriar Elahi Dhruvo, Mohammad Akhlaqur Rahman, Manash Kumar Mandal, Md Istiak Hossain Shihab, A. A. Ansary, Kaneez Fatema Shithi, Sanjida Khanom, **Rabeya Akter**, Safaeid Hossain Arib

RESEARCH EXPERIENCE

- **Research Assistant** March 2026 - Present
Electrical and Computer Engineering, North South University
Supervisor: [Md Salman Shamil](#) and [Dr. Md Adnan Arefeen](#)
 - Working on egocentric video understanding with a focus on modeling hand-object interactions and daily task actions with LLM-guided reasoning.
- **Research Assistant** Jun 2025 - Feb 2026
Department of Robotics and Mechatronics Engineering, University of Dhaka
Supervisor: [Dr. Md Mehedi Hasan](#) and [Md. Jubair Ahmed Surov](#)
 - Designed and implemented the first open-world amodal counting framework and assembled datasets to rigorously evaluate its performance.
 - Authored a manuscript detailing the methodology and empirical results, submitted to ECCV 2026.

■ Research Assistant

Jan 2023 - Dec 2023

Department of Robotics and Mechatronics Engineering, University of Dhaka

Supervisor: [Dr. Shafin Rahman](#) and [Dr. Sejuti Rahman](#)

- Designed and implemented a novel architecture that achieved competitive performance across Bangla, American, and German sign language benchmarks.
- Secured a competitive research grant from the Bangladesh Ministry of Science and Technology's Special Innovation Fund (Project ID: SRG-232431, Fiscal Year: 2023-2024).

INDUSTRY EXPERIENCE

■ Associate Software Engineer, QA

Apr 2024 - May 2025

[Therap \(BD\) Ltd., Dhaka, Bangladesh](#)

[\(Therap Services LLC, Connecticut, USA\)](#)

Team : Machine Learning

- Conducted research on Amazon Alexa devices and sound detection systems; performed literature review on action recognition datasets and contributed to the design of data collection for action recognition.
- Led quality assurance efforts for the video redaction system to develop and implement comprehensive test plans and cases to validate machine learning algorithms used in video processing.

■ AIM Intern

Jan 2024 - Mar 2024

[Pathao Limited, Dhaka, Bangladesh](#)

Team: Data Science, Analytics & Insights

- Developed 5 key performance indicators (KPIs) and metrics for brand health measurement of Pathao Courier, aligning with company-wide strategic goals.
- Designed and implemented comprehensive questionnaires using data-driven methodologies to assess brand health metrics, improving response quality and depth of insights.

TEACHING EXPERIENCE

■ Instructor

Jun 2024 - Aug 2024

National Camp, Bangladesh Artificial Intelligence Olympiad

- Designed and delivered lessons on classification algorithms and evaluation metrics for a group of high school students and created quizzes to assess their understanding.

SELECTED RESEARCH PROJECTS

■ Open-world Amodal Counting [\[GitHub\]](#)

Jun 2025 - Nov 2025

Supervisor: [Dr. Md Mehedi Hasan](#) and [Md. Jubair Ahmed Sourov](#)

Topic: Multimodal Learning, Representation Learning

- Addressed limitations in SOTA vision-language systems that fail to count objects under occlusion by developing CountOCC, the first open-world amodal counting framework.
- Co-developed the Feature Reconstruction Module (FRM) that infers class-discriminative features in occluded regions using spatial context and text-visual prompts, operating over a three-level feature pyramid.
- Designed Visual Equivalence (VisEQ) objective within a teacher-student distillation framework that enforces attention consistency between occluded and unoccluded views, improving generalization.
- Established occlusion-augmented evaluation protocols and demonstrated SOTA performance across structured and unstructured occlusion scenarios.

■ Language Guided Embodied Agents [\[GitHub\]](#)

May 2025 - Sep 2025

Topic: Vision Language Model, Embodied AI

- Contributed to developing the LaGEA framework that transforms VLM feedback (Qwen-2.5-VL-3B) into temporally localized dense rewards, enabling agents to understand not just what went wrong but why.
- Addressed credit assignment challenge in episodic VLM feedback by contributing to a reward pipeline that projects images, instructions, and feedback into a shared embedding space, measuring goal progress.
- Co-developed dynamic reward shaping strategy that provides strong guidance during early training and after failures, then gradually decays to prevent destabilizing training.
- Achieved SOTA with higher task success rates and faster convergence on Meta-World MT10 robotic manipulation benchmark.

■ Multivariate Time Series Forecasting [\[GitHub\]](#)

May 2025 - Sep 2025

Topic: Time Series Forecasting

- Co-developed T3Time tri-modal framework that learns shared representation space across temporal, spectral, and semantic modalities for multivariate time series forecasting.
- Identified that single-head cross-modal alignment failed to capture diverse temporal-spectral and semantic interactions across varying prediction horizons; designed adaptive multi-head cross-modal alignment module where multiple attention heads independently learn different interaction aspects.
- Led extensive experimental evaluation across 8 benchmark datasets, achieving SOTA performance with strong generalization.

■ Continuous Sign Language Translation [\[GitHub\]](#) [\[Demo\]](#) [\[Presentation\]](#)

Jan 2023 - Dec 2023

Topic: Computer Vision

- Identified through analysis that early Transformer models trained on video streams alone failed to capture sign-execution dynamics, the key factor distinguishing different signs, causing poor generalization.
- Led development of SignFormer-GCN, integrating video-stream Transformer with spatio-temporal GCN over skeletal keypoints to learn complementary appearance and motion representations, achieving SOTA performance across sign language benchmarks.

SELECTED ACADEMIC PROJECTS

- **Automated Stock Trading**[\[GitHub\]](#) Aug 2022 - Nov 2022
Supervisor: [Dr. Sejuti Rahman](#)
Topic: Reinforcement Learning
- **Human Activity Recognition Using Location Data** [\[GitHub\]](#) Apr 2021 - Jun 2021
Supervisor: [Dr. Md Atiqur Rahman Ahad](#)
Topic: Machine Learning
- **Comparative Analysis of Emotional Support Delivery by NAO Robot and Humans Across Varied Emotional States** [\[GitHub\]](#) [\[Demo\]](#) Aug 2023 - Nov 2023
Supervisor: [Dr. Lafifa Jamal](#)
Topic: Human-Robot Interaction

ACADEMIC SERVICE

- Reviewer at AAAI 2026

AWARDS AND SCHOLARSHIPS

- Engineering Faculty Undergraduate Merit Scholarship, University of Dhaka 2024
- Honourable Mention, NASA Earth Observatory (EO) Dashboard Hackathon [\[Link\]](#) 2021
- 11th, Sussex Huawei Locomotion Challenge 2021
- Second Runner-up, Ada Lovelace Datathon [\[Link\]](#) 2021
- 10th, 3rd Women's Mathematics Olympiad [\[Link\]](#) 2019
- 2nd, Pi Day Celebration [\[Link\]](#) 2019
- Dhaka Board Talent Pool Scholarship, Higher Secondary Certificate Examination 2018

LEADERSHIP/VOLUNTEER ACTIVITIES

- **Student Activity Secretary** Jan 2023 - Dec 2023
IEEE
University of Dhaka Student Branch
 - Organized several workshops, interactive sessions and industry expert experience sharing sessions.
- **Mentor** Jul 2019 - Dec 2020
Bangladesh Robot Olympiad (BdRO)
 - Conducted robotics workshops in multiple high schools and developed engaging quizzes for BdRO.
- **Speaker : Dui Prishthar Bisheshoggo** Dec 2023
IEEE Women In Engineering
 - Delivered a talk on personal experiences in hackathons and datathons, encouraging increased participation of women in tech competitions.